Vulnerability of small populations



A minimum viable population has to be of a sufficient size not only to maintain itself under average conditions but also to endure extreme events. Mark Shaffer likened this to a reservoir built to withstand the type of flood that occurs only once in 50 years, but not a devastating once-in-a-century flood.

Small populations are especially vulnerable to multiple threats occurring successively. The Heath Hen in New England, US, had been widespread in colonial times, but relentless hunting for food and sport caused a dramatic decline in Heath Hen numbers by 1908. In that year, the last surviving population on the island of Martha's Vineyard was given protected status. However, a catastrophic wildfire during the 1916 breeding season, severe winters, inbreeding, disease, and heavy predation by birds of prey all combined to push the Heath Hen population below a viable level. By 1927, only two females remained, and the species was extinct by 1932.

How studies are devised

PVA studies are now conducted in several ways. The simplest type is the time-series PVA, which looks at the entirety of a population over a period of time in order to calculate a rough average growth trend and any variations. In such studies, all individuals are treated as identical.

Demographic PVAs tend to be more precise and detailed. They are based on estimated reproductive and survival rates for different age bands within the population. Such analyses require much more data, but can provide